

This week you will begin by exploring the background of business analytics including its definitions, history, types and applications in various industries. You will also reflect on case studies to understand the application of business analytics and its success. You will read articles, watch videos and participate in a discussion to strengthen your foundational knowledge of business analytics.

Basic concepts of business analytics

The growth of business analytics

There has been tremendous growth in the field of business analytics and its application across industries and it continues to grow rapidly. The primary reason for the success of business analytics is its capability of driving business outcomes and improving revenue for organisations that apply it. This growth has also led to an increase in demand for talent who are well versed with the practice of business analytics as more and more companies start to implement it.

Here are some examples that confirm the growth of business analytics as a market and why it is important.

- According to a study by Gartner (2017), global revenue in the business intelligence (BI) and analytics market is forecast to reach \$18.3 billion in 2017, an increase of 7.3 per cent from 2016, and forecasted to grow to \$22.8 billion by 2020.
- Revenues for big data and business analytics will grow from \$130 billion by the end of 2016 to \$203 billion by 2020 – a compound growth rate of 11.7% (IDC, 2020).
- According to the *Harvard Business Review*, data scientist is the sexiest job of the twenty-first century (Davenport & Patil, 2012).



Read

Read the following articles to know more about the growing business analytics markets, future trends, and career opportunities.

- [*Gartner says worldwide business intelligence and analytics market to reach \\$18.3 billion in 2017*](#) (2017)
- [*6 predictions for the \\$203 billion big data analytics market*](#) (Press, 2017)
- [*25+ impressive big data statistics for 2022*](#) (Petrov, 2022)
- [*Data scientist: The sexiest job of the 21st century*](#) (Davenport & Patil, 2012)

The 'big data' phenomenon has changed the way businesses function. It has transformed the way businesses operate due to the amount and type of data being generated. So, what is big data?

“

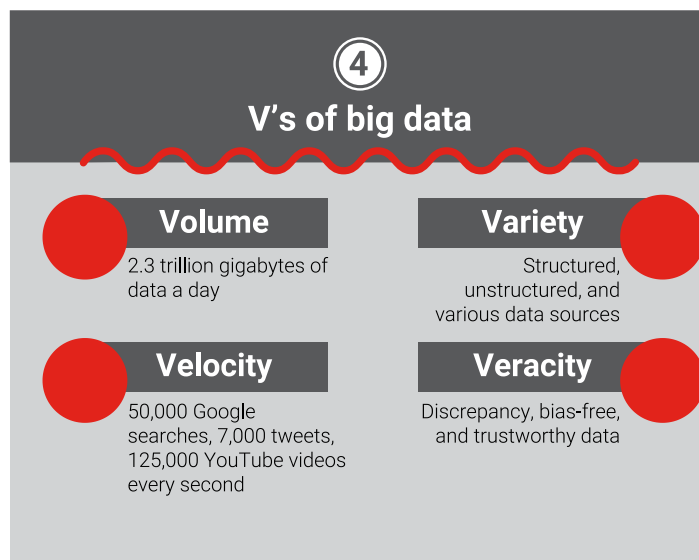
Big data is high-volume, -velocity and -variety information assets that demand cost-effective, innovative forms of information processing for enhanced insight and decision making.

”

(Gartner, 2022)

Figure 1.1

The four Vs of big data



(Click on the image to expand)

© Middea et al., 2015

[\[Image transcript\]](#)

The four Vs in the above definition can be described as:

- **Volume:** The amount of data being generated has now moved from terabytes to zettabytes.
- **Velocity:** The speed at which this data is being generated and processed is faster and links datasets at different frequencies and in real-time thanks to connected devices and social media.
- **Variety:** The data is being generated in different formats and sources such as structured/ unstructured, clickstream, text, emoji and multimedia.
- **Veracity:** When the data is easily available and in different formats, it raises concerns around the quality. This requires cleaning the data to be able to make sound business decisions.

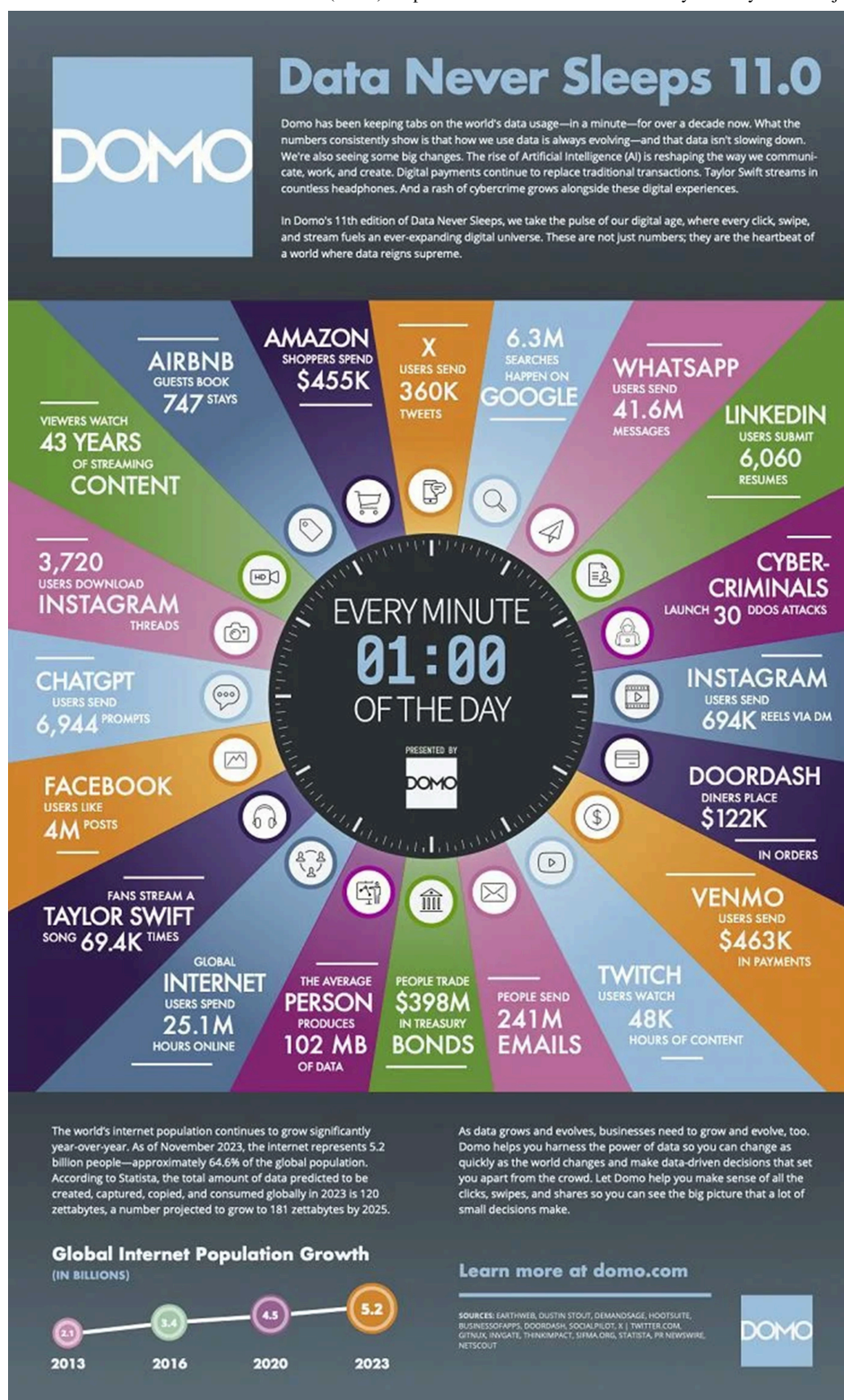
These changes in the way data is being generated and consumed has prompted businesses to apply business analytics to be able to understand their consumers and the environment they operate in.

The future of data

With more advanced technology becoming available, the trend of using data to drive business objectives is here to stay. Some key statistics regarding the future of data are captured in the image below.

Figure 1.2

Key data statistics



© Domo, 2023

[\[Image transcript\]](#)

According to this infographic:

- 500 million tweets are sent.
- 294 billion emails are sent.
- 4 petabytes of data are created on Facebook.
- 4 terabytes of data are created from each connected car (sensor data).
- 65 billion messages are sent on WhatsApp.
- 95 million images/videos are shared on Instagram; 5 billion searches are made.
- By 2025, it's estimated that 463 exabytes of data will be created each day globally.
- Most of the data is coming from the developing or Third World countries — streaming data generated from mobile and smart devices.

Introduction to business analytics

Business analytics involves processing the data generated to turn it into something useful and valuable that drives changes in the organisation to assist and aid strategic business decisions. As shown in the image below, the data is processed by people who are involved in cleaning and modelling it to generate insights or opportunities that prompt business decisions, which are then actioned to optimise business performance. This process could be triggered from both ends. It can be triggered from the data being generated and it can be triggered by an organisation's strategic business decisions, which triggers the analytics process.

Figure 1.3

The data analytics process



(Click on the image to expand)

© La Trobe University, 2024

[\[Image transcript\]](#)

One of the early definitions of business analytics that was presented in one of the analytics projects in the UK:

“

Business Analytics is the process of transforming data into insights that support, improve, and/or automate business decisions.

”

(The ORATER project, 2013)

This definition has since evolved and has been defined by the following organisations as shown:

“

- *Business analytics is comprised of solutions used to build analysis models and simulations to create scenarios, understand realities and predict future states.*
- *Business analytics includes data mining, predictive analytics, applied analytics and statistics, and is delivered as an application suitable for a business user.*

”

(Gartner, n.d)

“

Business analytics is a set of automated data analysis practices, tools and services that help you understand both what is happening in your business and why, to improve decision-making and help you plan for the future.

”

(IBM, n.d)

Holsapple et al. (2014) provided a more formal definition for business analytics. According to them, business analytics is:

- a movement — problem recognition and solving by using evidence
- a collection of practices and technologies — evidence without purpose (an organisation's acquisition, assimilation, selection, generation and emission of knowledge)
- a transformation process — focus on the process transforming evidence into insight

- a capability set — competencies possessed by an organization for transformation
- specific activities — activities based on available evidence instead of broad competencies
- a decisional paradigm — a model/approach to decision making (alongside / instead of experience, anecdotes and guesses).



Read

Read the following article to explore the Holsapple et al. (2014) definitions of business analytics in detail.

[*A unified foundation for business analytics*](#) (Holsapple et al., 2014)

Or listening to the below AI generated conversation (37 min.) to unpack this research article:



Read

Read the following article to explore some more definitions of business analytics.

[*Defining business analytics: An empirical approach*](#) (Power et al., 2018)

Types of analytics

The four most common types of analytics based on the organisation's strategic needs and requirements are:

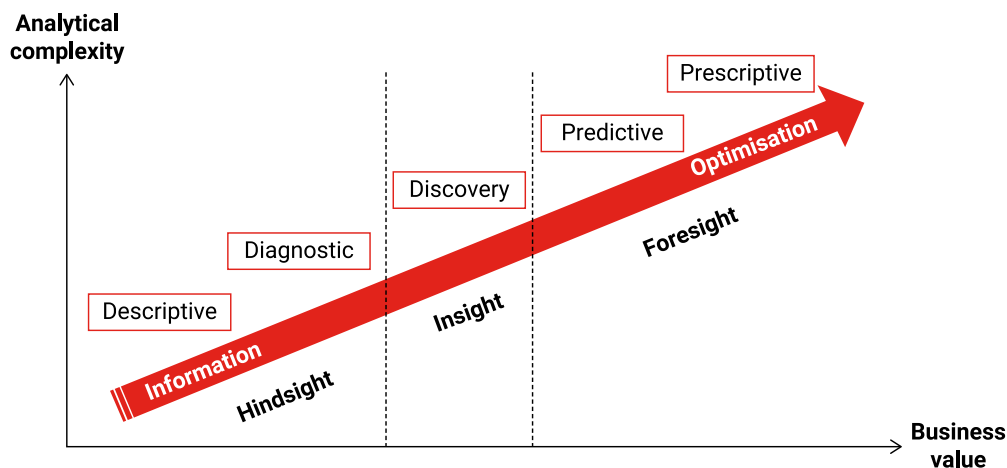
- **Descriptive analytics:** A basic, less complicated analytics that will describe what has happened so far. For example, descriptive analysis can be used to understand the sales performance of a company and if there has been a change due to the pandemic.
- **Diagnostic analytics:** Usually happens after descriptive analytics and tries to answer why certain events happened. For example, diagnostic analysis will identify why the sales in a particular period of the year were high. If there was a major festival, such as Christmas, did that lead to more sales?
- **Predictive analytics:** A more complex analytics to predict what might happen. This includes building predictive models to identify future trends based on historical data. For example, descriptive analytics can be used to identify the marketing campaigns that can be used to improve sales based on past performance.
- **Prescriptive analytics:** The final and one of the most challenging analytics that provides recommendations on what to do next. For example, prescriptive analytics can be used to understand the resources required to maximise profits.

There is another type of analytics that sits between descriptive and predictive analytics. It is called discovery analytics, or exploratory data analysis. Discovery or diagnostic analysis helps to understand the patterns in data through visualisation and identify anomalies, if any.

The process of conducting the three types of analytics is shown in the image below.

Figure 1.4

Three types of analytics



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[\[Image transcript\]](#)



Watch

Types of data analytics

This video [1.00 min.] is a quick recap of the types of analytics with examples. Watch this video to understand how different types of analytics can be applied in different scenarios.

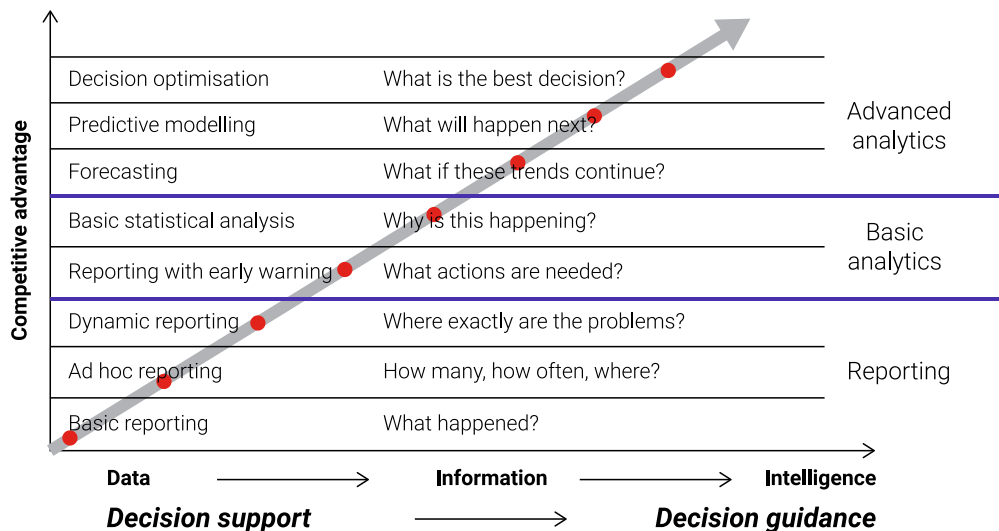


© 2017 by Dattabot Big Data Analytics. *3 Main Types of Analytics in Big Data*

The figure below gives another example of the analytics process. Note that as you move from basic analytics to complex analytics, it helps an organisation gain competitive advantage. Application of more advanced analytics improves the business values and outputs. The intelligence or insights gained by advanced analytics provide better predictions on business decisions.

Figure 1.5

Analytics and competitive advantage



(Click on the image to expand)

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[\[Image transcript\]](#)

Key terms in business analytics

Some of the key terms that you will come across in the practice of business analytics are:

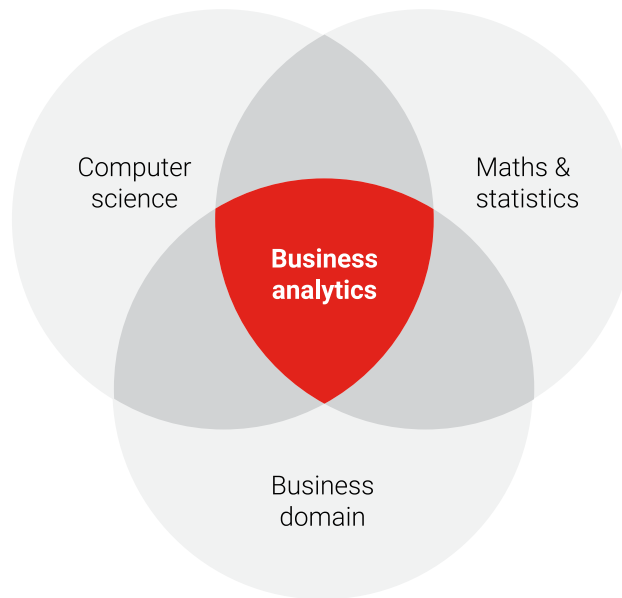
- business intelligence
- data science
- big data
- data mining

- knowledge discovery
- machine learning (and deep learning).

Business analytics covers different types of knowledge and skills as shown in the image below. It combines the practices of computer science, maths and statistics, and business domain knowledge. To be successful as a business analyst, it is important to have some background knowledge of these areas.

Figure 1.6

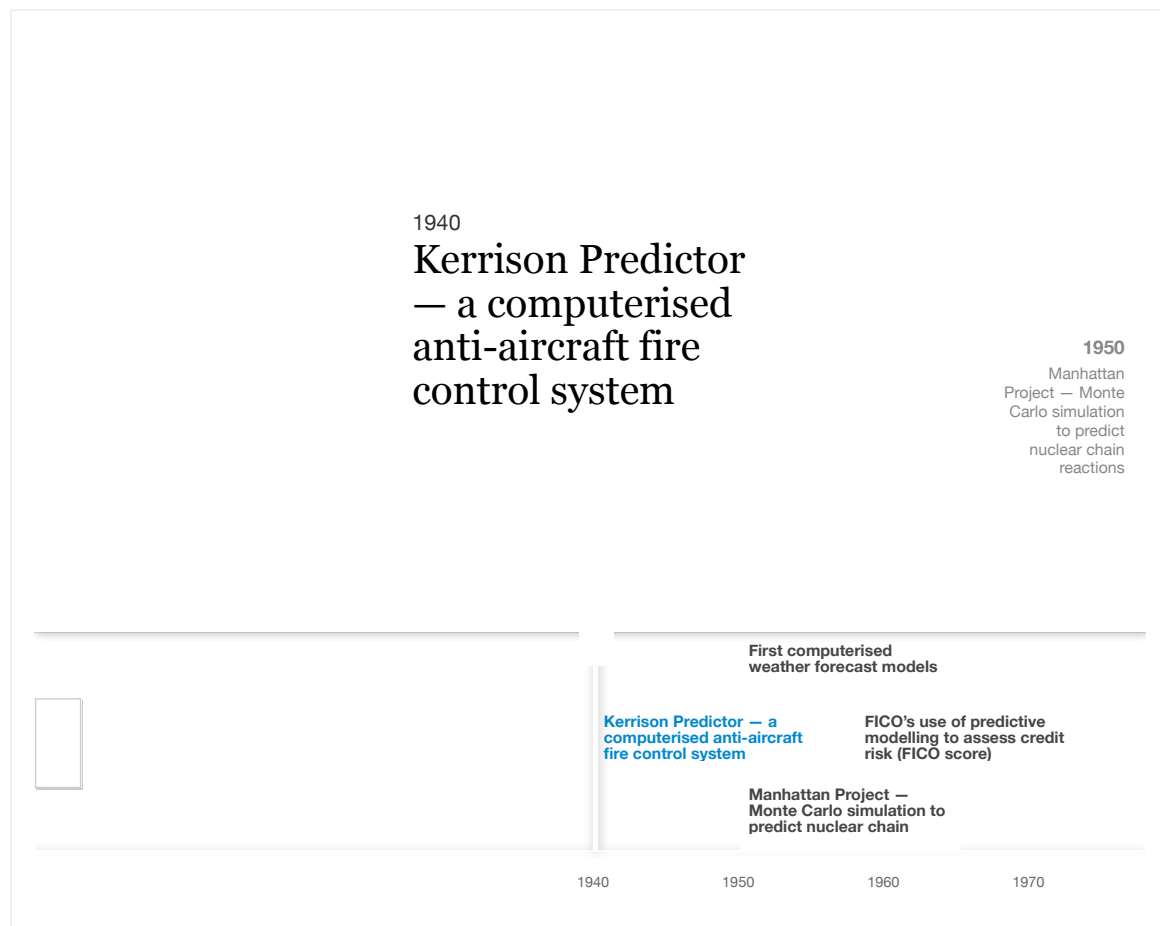
Business analytics domains



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History of analytics

Here is a brief timeline on the evolution of analytics and its application.



Business intelligence vs business analytics

Two of the most commonly used terms in the field of analytics are business intelligence (BI) and business analytics (BA). As an analyst, it is important for you to understand the difference between the two. Some of the major differences are:

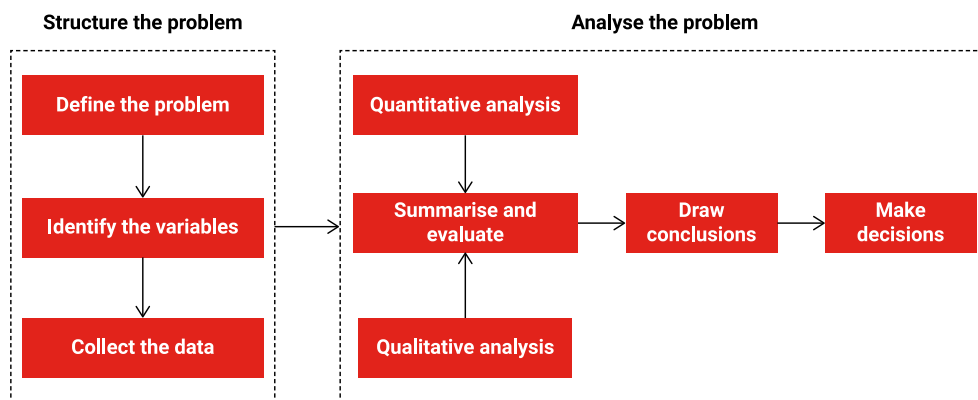
- BI mainly focuses on descriptive analytics.
- BI provides an overview of historical and present data to show what has happened or what is currently happening.
- BI answers the questions 'what' and 'how' so you can replicate what works and change what does not.
- BA focuses on predictive analytics.
- BA uses data mining, modelling, machine learning and deep learning to determine the likelihood of future outcomes and predictions.
- BA answers the question 'why' so it can provide predictions about what will happen.

An analytics approach

The methodology to set up an analytics project has two parts, as show in the image below.

Figure 1.7

Analytics methodology



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[Image transcript]

1. Structuring the problem — This part begins with defining the problem around the organisation's business goals, needs and strategies. Once the problem is defined, the next step is to identify the variables that we wanted to understand, their relationships and, finally, collect the relevant data. This could be internally housed within the organisation or obtained from external sources.
2. Analysing the problem — This involves deciding on the analysis methods/techniques, depending on the data types (quantitative/qualitative). The next step is to summarise and evaluate the analysis results. This should lead us to draw some conclusions (insights) and make some actionable decisions at the end.

This methodology is a generic approach to business analytics. We need a structured, reliable and repeatable approach that can be used as a standard framework for all the adopters (which might or might not have relevant knowledge) across various industry domains to better manage the analytics / data mining process. One of the common frameworks in the data mining space is the CRISP-DM framework. CRISP-DM stands for:

CRoss-Industry **S**tandard **P**rocess for **D**ata **M**ining (**CRISP-DM**)

Some other methodologies/frameworks used for data mining are KDD (Knowledge Discovery in Databases) and, SEMMA by SAS (Sample, Explore, Modify, Model and Access). In this subject, you will explore CRISP-DM in detail.

CRISP-DM Framework

CRISP-DM is an open standard data mining process model. It was conceived in 1996 by five leading European organisations in data mining, namely NCR Systems Engineering Copenhagen (United States and Denmark), DaimlerChrysler AG (Germany), SPSS Inc. (United States), and OHRA Verzekeringen en Bank Groep B.V. (The Netherlands). It was later acquired by IBM. IBM has incorporated CRISP-DM into its SPSS Modeler product (SPSS was acquired by IBM in 2009). CRISP-DM breaks the process of data mining into six major phases:

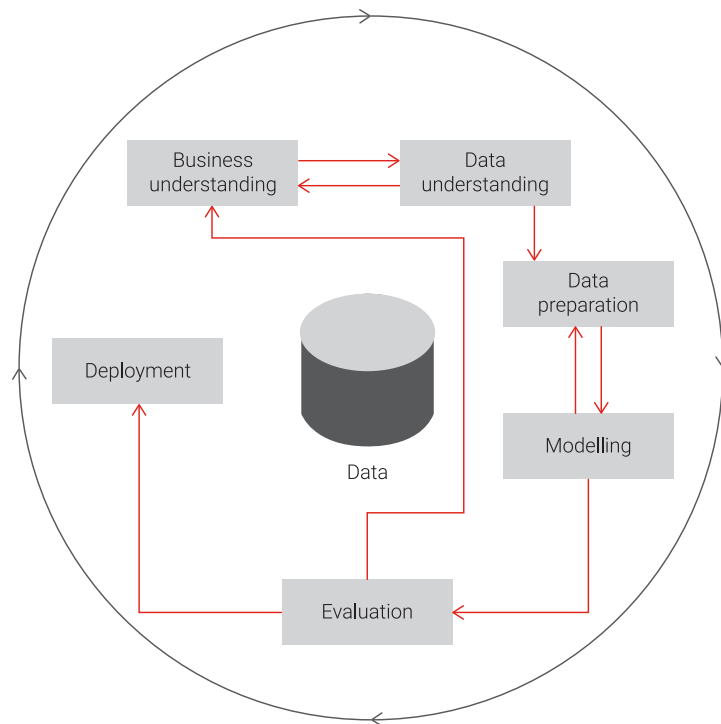
- business understanding
- data understanding
- data preparation

- modelling
- evaluation

The CRISP-DM framework is shown in the image below.

Figure 1.8

CRISP-DM framework



© Chapman et al., 2000

[\[Image transcript\]](#)

It is important to note that the sequence of the phases in this model is not rigid and can move back and forth between different phases as required. The outcome of each phase determines which phase, or which particular task of a phase, has to be performed next. The arrows in the process diagram indicate the most important and frequent 'dependencies' between these phases.

The main goals of the CRISP-DM framework are to:

- ensure quality of knowledge discovery project results
- reduce skills required for knowledge discovery
- reduce costs and time
- support documentation of projects
- capture experience for *reuse*
- support knowledge transfer and training.

When using CRISP-DM, it is important to remember that the framework should be general purpose (domain independent, application neutral) and robust (insensitive to environmental factors). Also, CRISP-DM should be supported by various tools and techniques, but should not be dependent.

Let's explore the phases of the CRISP-DM framework in detail.

Click on the below headings to reveal further information.

› **Business understanding**

› **Data understanding**

› **Data preparation**

› **Modelling**

› **Evaluation**

› **Deployment**

The outer circle in the diagram symbolises the cyclic nature of the data mining process itself. Data mining does not end once a solution is deployed. The data mining process continues after a solution has been deployed. The lessons learned during the process can trigger new, often more focused business questions and subsequent data mining processes will benefit from the experiences of previous ones. Regardless of whether the deployment is successful, the process often returns to the first phase (business understanding). The process of mining data produces a great deal of insights into the business problem and the difficulties of its solution. A second iteration can yield an improved model/solution. Just the experience of thinking about the business, the data and the performance goals often leads to new ideas for improving business performance, and even new lines of business or new ventures.

The table below provides an overview of the phases and tasks of the CRISP-DM model.

Figure 1.9

CRISP-DM phases and tasks

Business understanding	Data understanding	Data preparation	Modelling	Evaluation	Deployment
Determine business objectives Background Business objectives Business success criteria Assess situation Inventory of resources Requirements, assumptions, and constraints Risks and contingencies Terminology Costs and benefits Determine data mining goals Data mining goals Data mining success criteria Produce project plan Project plan Initial assessment of tools and techniques	Collect initial data Initial data collection report Describe data Data description report Explore data Data exploration report Verify data quality Data quality report	Select data Rationale for inclusion/exclusion Clean data Data cleaning report Construct data Derived attributes generated records Integrate data Merged data Format data Reformatted data Dataset Dataset description	Select modelling techniques Modelling technique Modelling assumptions Generate test design Test design Build model Parameter settings Models Model descriptions Assess model Model assessment Revised parameter settings	Evaluate results Assessment of data mining results w.r.t. business success criteria Approved models Review process Review of process Determine next steps List of possible actions decision	Plan deployment Deployment plan Plan monitoring and maintenance Monitoring and maintenance plan Produce final report Final report Final presentation Review project Experience documentation

(Click on the image to expand)

© Chapman et al., 2000

[\[Image transcript\]](#)

Determining data mining goals is an essential step when conducting data analytics. A data mining goal is different from a business goal in the sense that a business goal states objectives in business terminology, whereas a data mining goal states project objectives in technical terms. For example, the business goal might be — 'Increase catalogue sales to existing customers'. A data mining goal might be — 'Predict how many products a customer will buy, given their purchases over the past three years, relevant demographic information and the price of the product'.

Some of the activities involved in determining a business goal include:

- translating a business question to data mining goals. For example, a business goal is that a marketing campaign requires segmentation of customers to decide whom to approach in the campaign. This can be translated to a data mining goal, such as the level/size of the segments that need to be specified.
- specifying the data mining problem type. For example, classification, prediction, clustering etc.

Any model can be defined as shown in the image below where the input is data, the process is analytics and the output are the insights.

Figure 1.10

Defining a model



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All analytics can be represented as shown the image below.

Figure 1.11

Representation of analytics

$$\text{Outcome}_i = \text{Model}_i + \text{Error}_i$$

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A regression model is represented in the image below.

Figure 1.12

Regression model

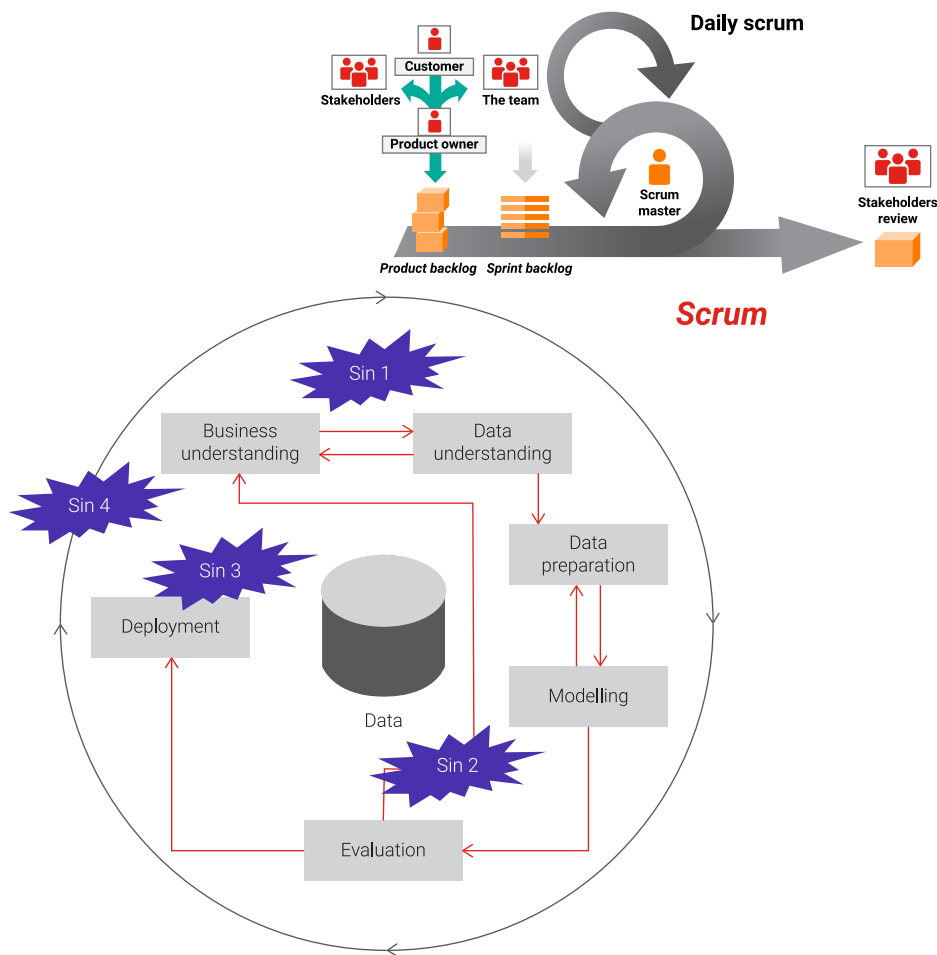
$$\text{Outcome}_i = (b_0 + b_1 \text{ Predictor}_i) + \text{Error}_i$$

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The CRISP-DM model also has its own set of challenges, often referred to as sins, as shown in the image below.

Figure 1.13

Challenges of CRISP-DM model



(Click on the image to expand)

© La Trobe University, 2024

[\[Image transcript\]](#)

Some of these sins are:

Click on the below headings to reveal further information.

> **Lack of clarity**

> **Data-driven iteration**

> **Blind hand-off to IT**

> **Poor adaptivity (failure to iterate)**

The best way to handle these sins is through analytics agility. Analytics agility is the capacity for business intelligence and analytics to be fast, responsive, adaptable and flexible.

Scrum, which is a subset of Agile, has been successfully implemented to resolve these issues. Some of the key features of Agile are:

- It is a lightweight process framework for agile development, and most widely used.
- A 'process framework' is a particular set of practices that must be followed for a process to be consistent with the framework. (For example, the Scrum process framework requires the use of development cycles called Sprints, the XP framework requires pair programming and so forth.)
- 'Lightweight' means that the overhead of the process is kept as small as possible — that is, maximise the amount of productive time available for getting useful work done.



Read

Read the following article to explore Agile in detail. Since it is a most commonly used tool in data analytics, having this knowledge will be beneficial for your career.

[*The definitive guide to what is Agile and Scrum*](#) (Agilemania, 2022)

Apart from CRISP-DM, there are some other approaches to analytics that have also been used. You can read more about them below.



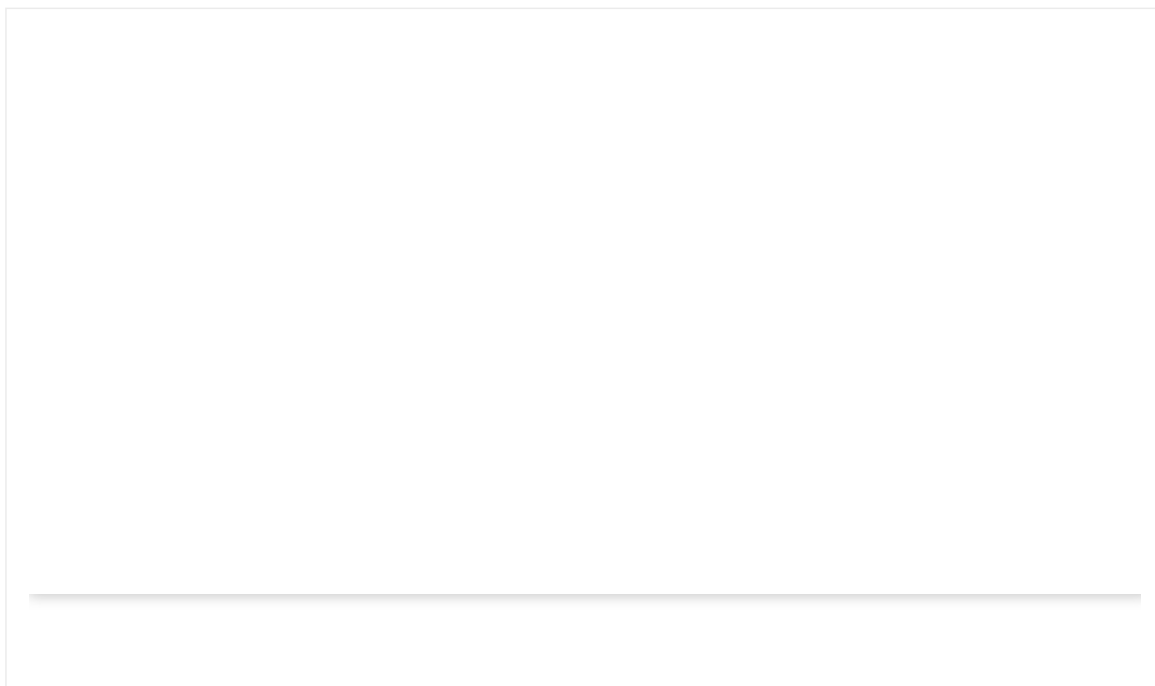
Read

Read the following articles to explore other analytics approaches in detail. Having this knowledge will be beneficial for your career.

- TDSP — Team data science process
[*What is the team data science process?*](#) (Microsoft, 2022)
- SAS analytics lifecycle
[*How well are you managing the analytical life cycle?*](#) (Moe, 2013)
- KNIME analytics lifecycle
[*The data science life cycle: A new standard for operationalizing data science*](#) (Berthold et al., 2021)

Success stories

Here are some examples of the successful application of business analytics across different sectors. These examples will help you understand the scope and depth of business analytics and the opportunities available for businesses.



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[Topic 2: Application of business analytics →](#)

